

Worksheet 1: Orbits and Formation

Full solutions

Covers Lectures 1 and 2: history of the field, planet formation, orbital dynamics

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

This worksheet is ungraded practice, with the same shape and level as the final exam. Plan up to 3 hours for a serious pass. Work each problem through on paper before you open the solutions, and show your algebra: the exam asks for the steps, not only the number. Some parts state an intermediate result ("show that..."); use it to check your work, and to continue if a step resists. Carry at least four significant figures through the intermediate steps (the worked solutions print five) and round only at the end.

Constants and data

$G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ $1 \text{ AU} = 1.496 \times 10^{11} \text{ m}$ $1 \text{ yr} = 3.156 \times 10^7 \text{ s}$ $1 \text{ d} = 86\,400 \text{ s}$
 $M_{\odot} = 1.989 \times 10^{30} \text{ kg}$ $M_{\oplus} = 5.972 \times 10^{24} \text{ kg}$ $R_{\oplus} = 6371 \text{ km}$

Body	a	P	Mass ($M_{\oplus}$)	ρ (kg m^{-3})
Mercury	0.387 AU	0.241 yr	0.055	5427
Venus	0.723 AU	0.615 yr	0.815	5243
Earth	1.000 AU	1.000 yr	1.000	5514
Mars	1.524 AU	1.881 yr	0.107	3934
Jupiter	5.203 AU	11.86 yr	317.8	1326
Saturn	9.537 AU	29.46 yr	95.16	687
Uranus	19.19 AU	84.01 yr	14.54	1271
Neptune	30.07 AU	164.8 yr	17.15	1638
Moon	$3.844 \times 10^5 \text{ km}$	27.32 d	0.0123	3344
Io		1.769 d		
Europa		3.551 d		
Ganymede	$1.0704 \times 10^6 \text{ km}$	7.155 d		

Saturn's mean radius is 58 232 km. Planetary and satellite values from NASA/JPL Solar System Dynamics; Galilean satellite periods as quoted in the Lecture 2 notes.

Problem 1 Weighing the Sun and Jupiter

Newton's form of Kepler's third law turns a period and an orbital size into a mass. Here you apply the same equation twice, to two different central bodies, and then ask how far it can be trusted.

(a) A planet of mass m moves on a circular orbit of radius r about a star of mass M , with $m \ll M$. Starting from the balance between the gravitational force and the centripetal force, derive

$$M = \frac{4\pi^2 r^3}{GP^2}.$$

Solution. The gravitational force supplies the centripetal force:

$$\frac{GMm}{r^2} = \frac{mv^2}{r}.$$

The orbital speed on a circular orbit is $v = 2\pi r/P$, so

$$\frac{GMm}{r^2} = \frac{m}{r} \left(\frac{2\pi r}{P} \right)^2 = \frac{m}{r} \frac{4\pi^2 r^2}{P^2} = \frac{4\pi^2 m r}{P^2}.$$

Divide both sides by m :

$$\frac{GM}{r^2} = \frac{4\pi^2 r}{P^2}.$$

Multiply both sides by r^2/G :

$$M = \frac{4\pi^2 r^3}{GP^2}.$$

(b) Judge this claim in 2 to 3 sentences: *"If Jupiter were twice as massive, its orbital period at its present distance would be noticeably longer."*

Solution. False, on two counts. The planet's mass cancels in part (a): it sets both the gravitational pull and the inertia, so the period contains only the stellar mass and the orbital size, and doubling Jupiter changes nothing at this order. In the exact two-body form, $P^2 = 4\pi^2 a^3 / [G(M + m)]$, a heavier planet raises the total attracting mass, so the period becomes slightly *shorter*, not longer; for a second Jupiter mass the change is $\frac{1}{2} M_J / M_\odot \approx 5 \times 10^{-4}$, far below "noticeable".

False: to leading order the planet's mass cancels, and the exact correction is tiny and has the opposite sign.

(c) Apply part (a) twice. First compute the mass of the Sun from Jupiter's orbit ($a = 5.203$ AU, $P = 11.86$ yr; show first that $r = 7.784 \times 10^{11}$ m and $P = 3.743 \times 10^8$ s). Then compute the mass of Jupiter from Ganymede's orbit ($a = 1.0704 \times 10^6$ km, $P = 7.155$ d; show that $P^2 = 3.822 \times 10^{11}$ s²), and express it in Earth masses.

Solution. The Sun from Jupiter. Convert to SI units:

$$r = 5.203 \times 1.496 \times 10^{11} \text{ m} = 7.7837 \times 10^{11} \text{ m}, \quad P = 11.86 \times 3.156 \times 10^7 \text{ s} = 3.7430 \times 10^8 \text{ s}.$$

Form the two powers:

$$r^3 = (7.7837 \times 10^{11})^3 = 4.7158 \times 10^{35} \text{ m}^3, \quad P^2 = (3.7430 \times 10^8)^2 = 1.4010 \times 10^{17} \text{ s}^2.$$

Numerator and denominator separately:

$$4\pi^2 r^3 = 39.478 \times 4.7158 \times 10^{35} = 1.8617 \times 10^{37} \text{ m}^3,$$

$$GP^2 = 6.674 \times 10^{-11} \times 1.4010 \times 10^{17} = 9.3503 \times 10^6 \text{ m}^3 \text{ kg}^{-1}.$$

Divide:

$$M = \frac{1.8617 \times 10^{37}}{9.3503 \times 10^6} = 1.991 \times 10^{30} \text{ kg}.$$

Jupiter from Ganymede. The same equation, with Jupiter now the central body:

$$a = 1.0704 \times 10^9 \text{ m}, \quad P = 7.155 \times 86\,400 \text{ s} = 6.1819 \times 10^5 \text{ s},$$

$$a^3 = (1.0704 \times 10^9)^3 = 1.2264 \times 10^{27} \text{ m}^3, \quad P^2 = (6.1819 \times 10^5)^2 = 3.8216 \times 10^{11} \text{ s}^2,$$

$$M = \frac{4\pi^2 a^3}{GP^2} = \frac{39.478 \times 1.2264 \times 10^{27}}{6.674 \times 10^{-11} \times 3.8216 \times 10^{11}} = \frac{4.8416 \times 10^{28}}{25.505} = 1.898 \times 10^{27} \text{ kg}.$$

In Earth masses:

$$\frac{1.898 \times 10^{27}}{5.972 \times 10^{24}} = 317.9.$$

| $M_{\odot} \approx 1.991 \times 10^{30} \text{ kg}$ (0.10% above the accepted $1.989 \times 10^{30} \text{ kg}$; **part (d) accounts for that excess**) and $M_J \approx 1.898 \times 10^{27} \text{ kg} = 317.9 M_{\oplus}$, **matching the data table.**

(d) Your solar mass in part (c) exceeds the accepted value by 0.10%. Use the exact form $P^2 = 4\pi^2 a^3 / [G(M+m)]$ to explain what that computation actually measured, and to account for the excess. Would the same effect be visible in your Jupiter mass? The mass of Ganymede is $1.482 \times 10^{23} \text{ kg}$.

Solution. The exact form returns the *sum* of the two masses, so the first computation, which neglected m , actually measured $M_{\odot} + M_J$:

$$M_{\odot} + M_J = M_{\odot} \left(1 + \frac{M_J}{M_{\odot}} \right), \quad \frac{M_J}{M_{\odot}} = \frac{1.898 \times 10^{27}}{1.991 \times 10^{30}} = 9.533 \times 10^{-4},$$

$$1.989 \times 10^{30} \times (1 + 9.533 \times 10^{-4}) = 1.9909 \times 10^{30} \text{ kg},$$

which agrees with the $1.9911 \times 10^{30} \text{ kg}$ of part (c) to 0.01%. The 0.10% excess is therefore not an arithmetic slip and not a rounding artefact: it is Jupiter's own mass. For Jupiter from Ganymede the same correction is

$$\frac{M_{\text{Gan}}}{M_J} = \frac{1.482 \times 10^{23}}{1.898 \times 10^{27}} = 7.81 \times 10^{-5},$$

that is 0.008%, a factor of 12 smaller and below the four-significant-figure precision of the input data.

| **The neglected orbiter mass shifts the derived central mass by the fraction m/M : 0.095% for Jupiter about the Sun, invisible 0.008% for Ganymede about Jupiter.**

Problem 2 The cheapest route to Mars

Treat the orbits of Earth and Mars as circular and coplanar, with radii 1.000 AU and 1.524 AU. A spacecraft transfers between them on an ellipse whose perihelion touches Earth's orbit and whose aphelion touches Mars's orbit. At each end the spacecraft fires its engine briefly (a *burn*), changing its speed by an increment Δv ; a burn along the direction of motion is *prograde*. Work entirely in the Sun's gravitational field and ignore the gravity of Earth and Mars themselves.

(a) Two spacecraft of equal mass are on orbits with the same semi-major axis a but different eccentricities. Judge each claim, with justification: (i) "*they have the same total energy*"; (ii) "*they pass through the distance $r = a$ at the same speed*".

Solution. Both claims are true. (i) The total energy of a Keplerian orbit is $E = -GMm/2a$, which depends on a alone; eccentricity only changes how the fixed total is divided

between kinetic and potential energy along the orbit. (ii) Put $r = a$ into the vis-viva equation:

$$v^2 = GM \left(\frac{2}{a} - \frac{1}{a} \right) = \frac{GM}{a}, \quad \text{so} \quad v = \sqrt{\frac{GM}{a}},$$

independent of e . On the circular orbit this speed holds everywhere; on the eccentric orbit it holds at the two points where the body is at distance a from the focus, the ends of the minor axis.

I Both true; both follow from E depending only on a .

(b) Show that the transfer ellipse has $a_t = 1.262$ AU. Explain why the flight time is half the period of that ellipse, and compute it in days.

Solution. The transfer ellipse has perihelion $r_p = 1.000$ AU and aphelion $r_a = 1.524$ AU. Since $r_p = a(1 - e)$ and $r_a = a(1 + e)$, adding the two gives

$$r_p + r_a = a(1 - e) + a(1 + e) = 2a,$$

so

$$a_t = \frac{r_p + r_a}{2} = \frac{1.000 + 1.524}{2} = 1.262 \text{ AU} = 1.88795 \times 10^{11} \text{ m}.$$

Departure is at perihelion and arrival is at aphelion. These are the two ends of the major axis, so they are separated by half a revolution and the flight takes half a period. With $GM = 6.674 \times 10^{-11} \times 1.989 \times 10^{30} = 1.32746 \times 10^{20} \text{ m}^3 \text{ s}^{-2}$:

$$a_t^3 = (1.88795 \times 10^{11})^3 = 6.7293 \times 10^{33} \text{ m}^3, \quad \frac{a_t^3}{GM} = \frac{6.7293 \times 10^{33}}{1.32746 \times 10^{20}} = 5.0693 \times 10^{13} \text{ s}^2,$$

$$P_t = 2\pi\sqrt{5.0693 \times 10^{13}} = 2\pi \times 7.1199 \times 10^6 = 4.4736 \times 10^7 \text{ s}.$$

Half of that is

$$t = \frac{1}{2}P_t = 2.2368 \times 10^7 \text{ s} = \frac{2.2368 \times 10^7}{86400} = 258.9 \text{ d}.$$

I $a_t = 1.262$ AU; flight time 259 d, about 8.5 months.

A faster route to the same number: dividing Kepler's third law for the transfer ellipse by the same law for Earth's orbit ($a = 1$ AU, $P = 1$ yr) cancels every constant and leaves $P^2 = a^3$ in units of AU and years, so $P_t = 1.262^{3/2} = 1.418$ yr and $t = 0.709$ yr.

(c) Use the vis-viva equation to show that the spacecraft moves at $v_p = 32.7 \text{ km s}^{-1}$ at the transfer perihelion and $v_a = 21.5 \text{ km s}^{-1}$ at the transfer aphelion. Then compute the circular orbital speeds of Earth and Mars, and from the four speeds the two velocity increments Δv_1 (departure), Δv_2 (arrival), and their sum.

Solution. At perihelion, $r = 1.000 \text{ AU} = 1.496 \times 10^{11} \text{ m}$ and $a = a_t = 1.88795 \times 10^{11} \text{ m}$:

$$\frac{2}{r} = 1.3369 \times 10^{-11} \text{ m}^{-1}, \quad \frac{1}{a_t} = 5.2968 \times 10^{-12} \text{ m}^{-1}, \quad \frac{2}{r} - \frac{1}{a_t} = 8.0722 \times 10^{-12} \text{ m}^{-1},$$

$$v_p^2 = GM \left(\frac{2}{r} - \frac{1}{a_t} \right) = 1.32746 \times 10^{20} \times 8.0722 \times 10^{-12} = 1.0716 \times 10^9 \text{ m}^2 \text{ s}^{-2},$$

$$v_p = 3.2735 \times 10^4 \text{ m s}^{-1} = 32.73 \text{ km s}^{-1}.$$

At aphelion, $r = 1.524 \text{ AU} = 2.2799 \times 10^{11} \text{ m}$:

$$\frac{2}{r} = 8.7723 \times 10^{-12} \text{ m}^{-1}, \quad \frac{2}{r} - \frac{1}{a_t} = 3.4755 \times 10^{-12} \text{ m}^{-1},$$

$$v_a^2 = 1.32746 \times 10^{20} \times 3.4755 \times 10^{-12} = 4.6136 \times 10^8 \text{ m}^2 \text{ s}^{-2},$$

$$v_a = 2.1479 \times 10^4 \text{ m s}^{-1} = 21.48 \text{ km s}^{-1}.$$

The circular speeds are the $r = a$ case of the vis-viva equation:

$$v_{\oplus} = \sqrt{\frac{GM}{r_p}} = \sqrt{\frac{1.32746 \times 10^{20}}{1.496 \times 10^{11}}} = \sqrt{8.8734 \times 10^8} = 29.79 \text{ km s}^{-1},$$

$$v_{\text{Mars}} = \sqrt{\frac{1.32746 \times 10^{20}}{2.2799 \times 10^{11}}} = \sqrt{5.8224 \times 10^8} = 24.13 \text{ km s}^{-1}.$$

The increments:

$$\Delta v_1 = v_p - v_{\oplus} = 32.73 - 29.79 = 2.95 \text{ km s}^{-1},$$

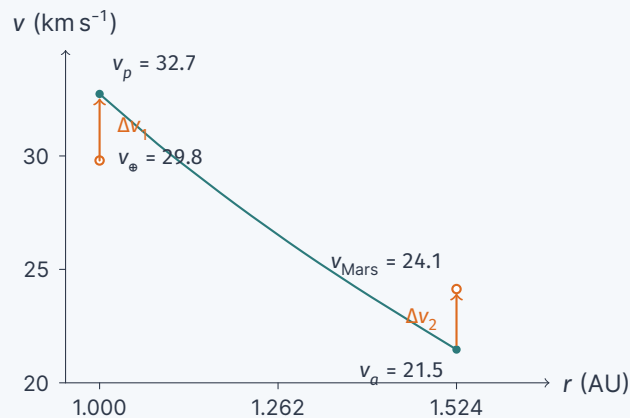
$$\Delta v_2 = v_{\text{Mars}} - v_a = 24.13 - 21.48 = 2.65 \text{ km s}^{-1},$$

$$\Delta v_{\text{total}} = 2.95 + 2.65 = 5.60 \text{ km s}^{-1}.$$

■ $\Delta v_1 = 2.95$, $\Delta v_2 = 2.65$, **total** $\Delta v = 5.60 \text{ km s}^{-1}$.

(d) Sketch the spacecraft's speed $v(r)$ along the transfer, from departure at 1.000 AU to arrival at 1.524 AU. Mark the four speeds of part (c): v_p , v_a , and the circular speeds of Earth and Mars. Use the sketch to explain why *both* burns point in the direction of motion.

Solution. The vis-viva speed falls monotonically from perihelion to aphelion:



At departure the transfer demands 32.7 km s^{-1} while Earth's orbit provides 29.8 : the spacecraft must *gain* speed, so the first burn is prograde. At arrival the spacecraft crawls at its aphelion speed of 21.5 km s^{-1} while Mars passes at 24.1 : the spacecraft is the slower body and must again *gain* speed to match, so the second burn is prograde too.

■ **The transfer curve starts above Earth's circular speed and ends below Mars's; closing each gap means adding speed along the direction of motion, so both burns are prograde.**

Problem 3 The Laplace resonance of the Galilean moons

Lecture 2 describes the orbital periods of Io, Europa and Ganymede as standing in a 1 : 2 : 4 ratio. This problem asks how exact that statement is, and what the resonance geometry enforces.

(a) Compute the two consecutive period ratios and the Ganymede-to-Io ratio, and give the percentage by which each misses its integer value.

Solution.

$$\begin{aligned}\frac{P_{\text{Eur}}}{P_{\text{Io}}} &= \frac{3.551}{1.769} = 2.00735, & \frac{2.00735 - 2}{2} &= +0.37\%, \\ \frac{P_{\text{Gan}}}{P_{\text{Eur}}} &= \frac{7.155}{3.551} = 2.01493, & \frac{2.01493 - 2}{2} &= +0.75\%, \\ \frac{P_{\text{Gan}}}{P_{\text{Io}}} &= \frac{7.155}{1.769} = 4.04466, & \frac{4.04466 - 4}{4} &= +1.12\%.\end{aligned}$$

The ratios exceed 2, 2 and 4 by 0.37%, 0.75% and 1.12%.

These offsets are real, not measurement error: the periods are known to far better than a part in 10^4 . Part (b) shows what the resonance does hold exactly: a combination of all three mean motions, not any pairwise ratio.

(b) The mean motion is $n = 2\pi/P$. Evaluate the combination

$$n_{\text{Io}} - 3n_{\text{Eur}} + 2n_{\text{Gan}}$$

and express the result as a fraction of n_{Io} . Keep 7 decimal digits through the arithmetic; as a checkpoint, $3n_{\text{Eur}} = 0.8448324 \times 2\pi \text{ d}^{-1}$. Contrast the outcome with part (a).

Solution. The factor 2π is common to all three terms, so it can be carried outside and the arithmetic done with $1/P$ in units of d^{-1} :

$$\begin{aligned}\frac{1}{P_{\text{Io}}} &= \frac{1}{1.769} = 0.5652911 \text{ d}^{-1}, \\ \frac{3}{P_{\text{Eur}}} &= \frac{3}{3.551} = 3 \times 0.2816108 = 0.8448324 \text{ d}^{-1}, \\ \frac{2}{P_{\text{Gan}}} &= \frac{2}{7.155} = 2 \times 0.1397624 = 0.2795248 \text{ d}^{-1}.\end{aligned}$$

Combine, keeping every digit because this is a difference of nearly equal numbers:

$$\begin{aligned}0.5652911 - 0.8448324 &= -0.2795413 \text{ d}^{-1}, \\ -0.2795413 + 0.2795248 &= -0.0000165 \text{ d}^{-1},\end{aligned}$$

so, restoring the 2π ,

$$n_{\text{Io}} - 3n_{\text{Eur}} + 2n_{\text{Gan}} = -1.04 \times 10^{-4} \text{ rad d}^{-1}, \quad \frac{n_{\text{Io}} - 3n_{\text{Eur}} + 2n_{\text{Gan}}}{n_{\text{Io}}} = -2.9 \times 10^{-5}.$$

The combination vanishes to three parts in 10^5 , although no individual ratio in part (a) came closer than three parts in 10^3 .

This is the Laplace relation. Each pairwise ratio misses its integer by a few tenths of a percent, yet the particular three-body combination is zero to a hundred times better precision. The exact constraint the system obeys is the one on all three moons together, not the one on any pair. With the NASA/JPL sidereal periods quoted to seven digits (1.769138, 3.551181, 7.154553 d) the residual falls further, to about 10^{-7} of n_{Io} .

(c) The figure shows the three orbits at the moment of a Europa–Ganymede conjunction; the resonance locks the phases such that Io then stands on the *opposite* side of Jupiter.



Idealise the periods as exactly 1 : 2 : 4. Sketch the positions of the three moons after 1, 2, and 4 Io periods. From your sketch: (i) where does Io stand at every Europa–Ganymede conjunction, and (ii) can a triple conjunction occur?

Solution. Per Io period, Io advances 360° , Europa 180° , and Ganymede 90° . Starting from the printed configuration (Io at 180° , Europa and Ganymede at 0°):

Time (Io periods)	Io	Europa	Ganymede
0	180°	0°	0°
1	180°	180°	90°
2	180°	0°	180°
4	180°	0°	0°



Because Io always advances by full turns per Io period, it returns to 180° at every integer time; Europa alternates between 0° and 180° ; Ganymede steps through 90° per Io period. The pattern repeats after 4 Io periods, one Ganymede period.

(i) Europa–Ganymede conjunctions occur at $t = 0, 4, 8, \dots$, always at 0° , and Io then stands at 180° : at every Europa–Ganymede conjunction, Io is on the opposite side of Jupiter. (ii) Each pairwise conjunction finds the third moon elsewhere: Io–Europa conjunctions (odd t , at 180°) find Ganymede at 90° or 270° ; the Io–Ganymede conjunction ($t = 2$, at 180°) finds Europa at 0° .

A triple conjunction never occurs. The resonance phase-protects the system: the moons never line up all on one side, which caps how coherently their mutual perturbations can add.

(d) The resonance holds Io's eccentricity at $e \approx 0.004$ rather than letting tides circularise

the orbit. Explain why a non-zero eccentricity is needed for tidal heating to continue in a moon that is already tidally locked, and name the observable consequence.

Solution. A tidally locked moon on a perfectly circular orbit keeps the same face toward the planet at a constant distance. Its tidal bulge is then fixed in the body frame: it neither rises and falls nor moves across the surface, so no work is done deforming the interior and no heat is released.

An eccentric orbit breaks this in two ways at once. The distance to the planet varies over each orbit, so the height of the bulge varies with it. And the moon spins at a constant rate while its orbital angular velocity is fastest at periapsis and slowest at apoapsis, so the direction to the planet librates back and forth in longitude and drags the bulge with it. Both effects flex the interior once per orbit, and internal friction converts that mechanical work into heat.

Left alone, tidal dissipation would damp the eccentricity to zero and switch itself off. The Laplace resonance prevents that by continually re-forcing e .

Observable consequence: Io is the most volcanically active body in the solar system.

Problem 4 Tides and the Roche limit

(a) A moon of radius R_s orbits at distance d from a planet of mass M_p , with $R_s \ll d$. Show that the difference between the planet's gravitational acceleration at the moon's near surface and at its centre is

$$\Delta a \approx \frac{2GM_p R_s}{d^3}.$$

Use $(1 - x)^{-2} \approx 1 + 2x$ for $x \ll 1$.

Solution. The near surface sits at distance $d - R_s$ from the planet, the centre at d :

$$a_{\text{near}} = \frac{GM_p}{(d - R_s)^2}, \quad a_{\text{centre}} = \frac{GM_p}{d^2}.$$

Factor d out of the first denominator:

$$a_{\text{near}} = \frac{GM_p}{d^2} \left(1 - \frac{R_s}{d}\right)^{-2} \approx \frac{GM_p}{d^2} \left(1 + \frac{2R_s}{d}\right).$$

Subtract:

$$\Delta a = a_{\text{near}} - a_{\text{centre}} = \frac{GM_p}{d^2} \left(1 + \frac{2R_s}{d}\right) - \frac{GM_p}{d^2} = \frac{GM_p}{d^2} \cdot \frac{2R_s}{d} = \frac{2GM_p R_s}{d^3}.$$

The direct pull falls as d^{-2} but the tidal difference falls as d^{-3} , so their ratio is $\Delta a/a = 2R_s/d$: tides are a small correction far away and dominate close in.

(b) The fluid Roche limit is

$$d_R \approx 2.46 R_p \left(\frac{\rho_p}{\rho_s}\right)^{1/3}.$$

Interrogate this formula before using it: (i) why does the *satellite's* radius R_s not appear? (ii) what does it predict as $\rho_s \rightarrow \infty$, and does that make sense? (iii) what does it predict for

$\rho_s = \rho_p$, and what does that say about how close *any* self-gravitating satellite can orbit?

Solution. (i) Both sides of the comparison scale the same way with satellite size: the tidal acceleration across the satellite, $2GM_p R_s / d^3$, grows linearly with R_s , and so does the satellite's own surface gravity, $Gm_s / R_s^2 = \frac{4}{3}\pi G \rho_s R_s$. The radius cancels in the ratio and only the densities survive. Big and small satellites of the same density are torn apart at the same distance.

(ii) As $\rho_s \rightarrow \infty$, $d_R \rightarrow 0$: an arbitrarily dense satellite can orbit arbitrarily close, because its self-gravity grows without bound while the tidal stress at fixed d does not. The trend is physically sensible.

(iii) For $\rho_s = \rho_p$ the density factor is unity and $d_R = 2.46 R_p$: even a satellite as dense as its planet is destroyed within 2.46 planetary radii. The limit falls inside the planet, $d_R < R_p$, only for $\rho_s > 2.46^3 \rho_p \approx 15 \rho_p$, far denser than any satellite material, so for every real satellite density the forbidden zone extends well beyond the planet's surface.

R_s cancels because tidal stress and self-gravity both scale linearly with it; only the density ratio matters, and even equal density leaves a forbidden zone of $2.46 R_p$.

(c) Evaluate d_R for Saturn ($R_p = 58\,232$ km, $\rho_p = 687$ kg m⁻³) and a satellite of solid water ice ($\rho_s = 1000$ kg m⁻³); as a checkpoint, $(687/1000)^{1/3} = 0.8824$. Saturn's main rings span 67 000 to 137 000 km from the planet's centre, with the F ring near 140 000 km: compare. Ring particles are in fact porous aggregates with $\rho_s \approx 500$ kg m⁻³; *without* recomputing from scratch, state the factor by which the limit moves and what that does to your comparison.

Solution. Solid ice:

$$d_R = 2.46 \times 58\,232 \times 0.8824 = 2.46 \times 51\,384 = 1.264 \times 10^5 \text{ km} = 2.17 R_p.$$

This sits at 126 400 km: the C, B and inner A rings lie inside it, but the outer A ring (126 400 to 137 000 km) and the F ring lie outside, which the solid-ice limit fails to explain.

Halving the density multiplies d_R by $(1000/500)^{1/3} = 2^{1/3} = 1.26$, because $d_R \propto \rho_s^{-1/3}$:

$$d_R = 1.26 \times 126\,400 = 1.59 \times 10^5 \text{ km} = 2.73 R_p,$$

and the *entire* ring system, F ring included, now lies inside the limit.

$d_R = 126\,400$ km ($2.17 R_p$) for solid ice; porosity moves it out by $2^{1/3} = 1.26$ to $159\,000$ km ($2.73 R_p$), enclosing the whole ring system.

(d) For the Earth–Moon pair the same formula gives $d_R \approx 18\,500$ km $= 2.91 R_\oplus$; the Moon orbits 21 times further out. The International Space Station orbits 420 km above Earth's surface, far *inside* that limit. Judge this claim in 2 to 3 sentences: *"the ISS survives inside the Roche limit, so the Roche formula must be wrong."*

Solution. False. The Roche limit compares the tidal stress with a body's *self-gravity*, so it applies only to objects held together by gravity alone: fluid bodies and loose aggregates bound only by their own self-gravity. The ISS is held together by the material strength of its structure, which at its mass exceeds its self-gravity by many orders of magnitude, so it sits outside the formula's domain rather than refuting it.

The same escape applies to solid rock, which is why small rigid moons survive well inside the fluid limit.

False: the formula presupposes self-gravity as the only glue; the ISS is bolted, not gravitationally bound.

Problem 5 From planetesimals to planets, and the mass of the solar nebula

(a) Gravitational focusing raises the collision cross-section of a body of radius R to

$$\sigma_{\text{eff}} = \pi R^2 \left(1 + \frac{v_{\text{esc}}^2}{v_{\infty}^2} \right).$$

Two planetesimals of density 3000 kg m^{-3} have radii 50 km and 500 km. Show that the smaller body has $v_{\text{esc}} = 64.8 \text{ m s}^{-1}$, and explain why $v_{\text{esc}} \propto R$ at fixed density. Then compute both focusing factors for a swarm with random velocity $v_{\infty} = 10 \text{ m s}^{-1}$, and compare the ratio of the two effective cross-sections with the ratio of the geometric ones.

Solution. For the 50 km body, $R = 5.0 \times 10^4 \text{ m}$:

$$M = \frac{4}{3}\pi R^3 \rho = \frac{4}{3}\pi (5.0 \times 10^4)^3 \times 3000 = 1.571 \times 10^{18} \text{ kg},$$

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2 \times 6.674 \times 10^{-11} \times 1.571 \times 10^{18}}{5.0 \times 10^4}} = \sqrt{4194} = 64.8 \text{ m s}^{-1}.$$

At fixed density $M \propto R^3$, so $v_{\text{esc}} = \sqrt{2GM/R} \propto \sqrt{R^3/R} = R$: the 500 km body has ten times the escape speed, $v_{\text{esc}} = 648 \text{ m s}^{-1}$.

Focusing factors at $v_{\infty} = 10 \text{ m s}^{-1}$, using $v_{\text{esc}}^2 = 4194 \text{ m}^2 \text{ s}^{-2}$ from above and, since $v_{\text{esc}} \propto R$, 100 times that value for the larger body:

$$1 + \frac{4194}{10^2} = 1 + 41.9 = 42.9 \quad \text{and} \quad 1 + \frac{4.194 \times 10^5}{10^2} = 1 + 4194 = 4195.$$

The cross-sections are in the ratio

$$\frac{\sigma_{\text{eff}}(500)}{\sigma_{\text{eff}}(50)} = \left(\frac{500}{50} \right)^2 \times \frac{4195}{42.9} = 100 \times 97.8 = 9.8 \times 10^3,$$

against a geometric ratio of only $(500/50)^2 = 100$.

Focusing gives the larger body an extra factor of 98 on top of its geometric advantage.

(b) Read the limits of the focusing formula: what does σ_{eff} reduce to for $v_{\infty} \gg v_{\text{esc}}$, and how does it scale with R for $v_{\infty} \ll v_{\text{esc}}$? Confirm the first limit by evaluating both focusing factors at $v_{\infty} = 500 \text{ m s}^{-1}$. The growing bodies themselves stir the swarm toward their own escape speed: name the two growth regimes these limits correspond to, and explain why the first regime shuts itself down.

Solution. For $v_{\infty} \gg v_{\text{esc}}$ the bracket tends to 1 and $\sigma_{\text{eff}} \rightarrow \pi R^2$: the purely geometric

cross-section, growing only as R^2 . For $v_\infty \ll v_{\text{esc}}$ the second term dominates,

$$\sigma_{\text{eff}} \approx \pi R^2 \frac{v_{\text{esc}}^2}{v_\infty^2} \propto R^2 \times R^2 = R^4,$$

using $v_{\text{esc}} \propto R$ from part (a): the biggest body gains a steeply super-geometric advantage.

At $v_\infty = 500 \text{ m s}^{-1}$, with the same v_{esc}^2 values:

$$1 + \frac{4194}{500^2} = 1.02, \quad 1 + \frac{4.194 \times 10^5}{500^2} = 2.68,$$

so the large body's advantage over pure R^2 scaling collapses from the factor 98 of part (a) to 2.6.

The cold-swarm limit is **runaway growth**: the largest body eats fastest, pulls away from the swarm, and the mass spread widens. The hot-swarm limit is **oligarchic growth**: a few similar-sized bodies grow slowly and in step. The transition is self-inflicted: the winners of runaway growth gravitationally stir the planetesimals, raising v_∞ toward their own v_{esc} , which switches their focusing advantage off.

Geometric πR^2 for a hot swarm, $\propto R^4$ for a cold one; runaway growth creates the stirring that ends it, handing over to oligarchic growth.

(c) Estimate the minimum mass of the solar nebula. Assume the four terrestrial planets are made entirely of solids, each of the four giant planets needed a solid core of $15 M_\oplus$, and solids make up 1% of the disk by mass. As a checkpoint, the terrestrial planets contribute $1.98 M_\oplus$ of solids. Give the answer in Earth masses and as a fraction of $M_\odot$.

Solution. Solids locked into the terrestrial planets:

$$0.055 + 0.815 + 1.000 + 0.107 = 1.977 M_\oplus.$$

Solids locked into the four giant-planet cores:

$$4 \times 15 = 60 M_\oplus.$$

Total solid inventory:

$$M_{\text{solid}} = 1.977 + 60 = 61.98 M_\oplus.$$

If solids are 1% of the disk by mass, the disk must have held

$$M_{\text{disk}} = \frac{61.98}{0.01} = 6198 M_\oplus = 6198 \times 5.972 \times 10^{24} = 3.701 \times 10^{28} \text{ kg},$$

$$\frac{M_{\text{disk}}}{M_\odot} = \frac{3.701 \times 10^{28}}{1.989 \times 10^{30}} = 0.0186.$$

About $6.2 \times 10^3 M_\oplus$, or 1.9% of a solar mass.

This is a lower bound twice over. Giant-planet heavy-element inventories are probably larger than a bare $15 M_\oplus$ core, and no disk converts 100% of its solids into planets, so the true requirement is higher.

(d) By the Class II stage of disk evolution, a typical disk holds about 0.1% of the stellar mass. Judge this claim, with numbers: "a Class II disk still holds enough material to

build the solar system's planets." What does your verdict imply for *when* the planets' solid material had to be locked into large bodies?

Solution. A Class II disk around the Sun holds

$$0.001 \times \frac{1.989 \times 10^{30}}{5.972 \times 10^{24}} = 333 M_{\oplus}$$

of gas and dust in total. The claim fails on both budgets. The four giant planets alone sum to

$$317.8 + 95.16 + 14.54 + 17.15 = 444.7 M_{\oplus},$$

more than the entire disk. And at 1% dust, the disk carries only $3.33 M_{\oplus}$ of solids against the $61.98 M_{\oplus}$ required by part (c), short by a factor of

$$\frac{61.98}{3.33} = 18.6.$$

False: a Class II disk holds neither the gas ($333 < 445 M_{\oplus}$) nor the solids (3.3 vs $62 M_{\oplus}$) that the planets demand.

The implication is about timing. The solid inventory, and the giant-planet cores with their gas envelopes, had to be assembled *earlier*, while the disk was still young and held a few percent of a solar mass, as in part (c). This is consistent with giant-planet cores completing their growth within the first few Myr, before the gas disperses.